



National Forensic Sciences University, Gandhinagar

School of Cyber Security and Digital Forensics

Assignment: Practical - 6 : Wireshark

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Sem: 2

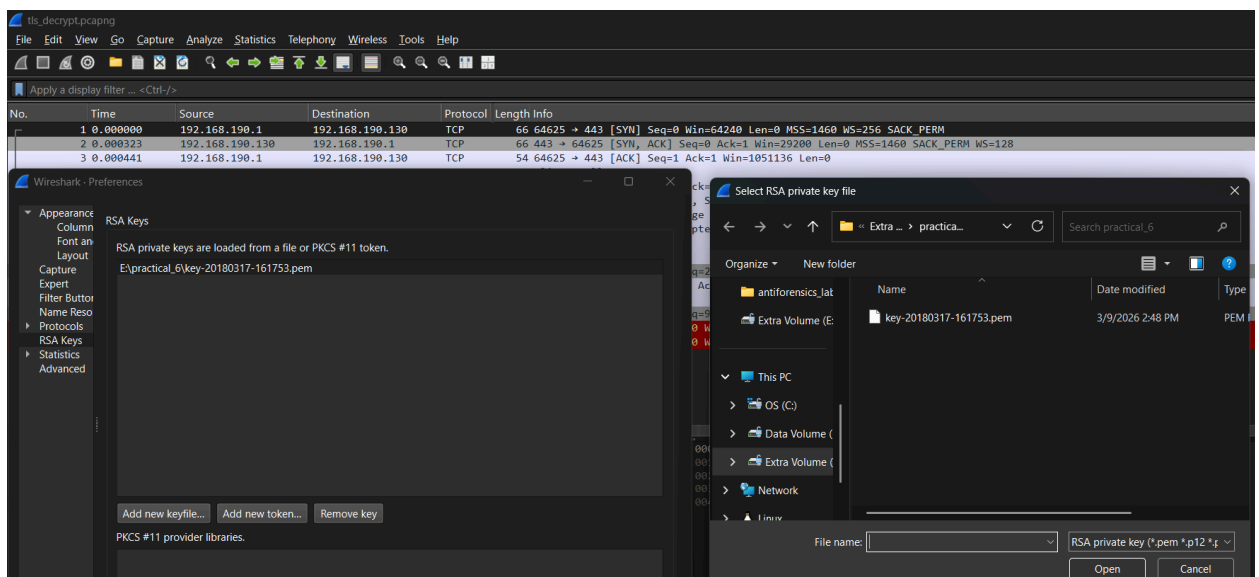
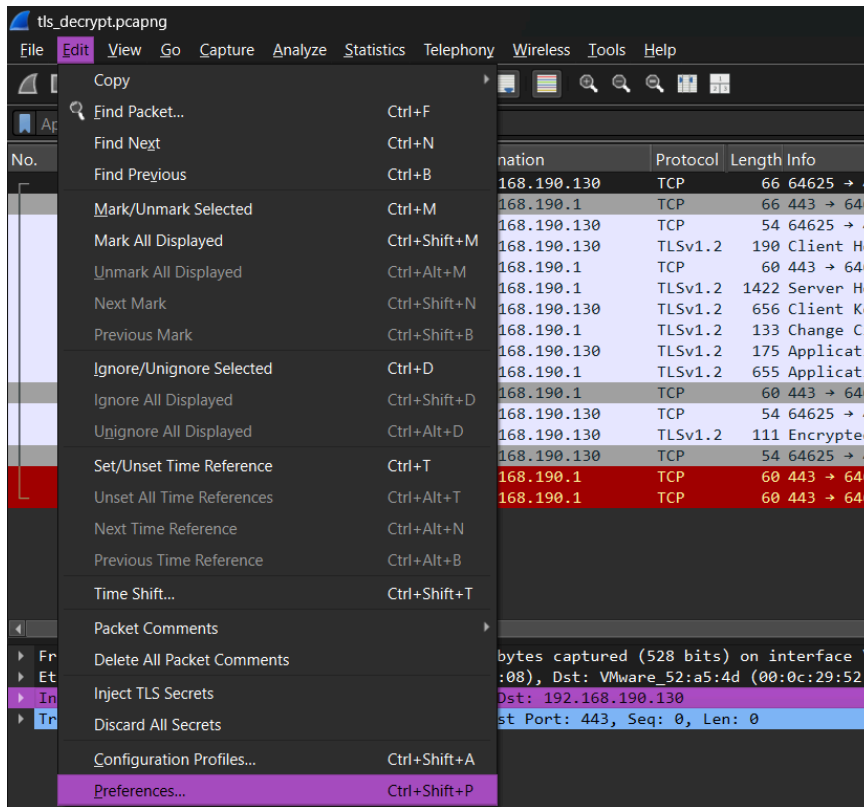
Subject: Network Security

Course: MSc Cyber Security

Batch: 2025 -2027

Tls_decrypt.pcapng

1. Configure Wireshark to decrypt the HTTPS traffic using the provided private key (key-20180317-161753.pem). Ans: Go to Edit > Preferences > Protocols > TLS. In the RSA keys list, add the provided private key file (key-20180317-161753.pem).



2. After decrypting the HTTP response, what message is displayed on the web Page? Ans: Welcome Home was displayed. Ans: Filter for http to find the decrypted traffic.

Right-click a decrypted HTTP packet and select Follow > HTTP Stream.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.190.1	192.168.190.130	TCP	66	64625 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.000323	192.168.190.130	192.168.190.1	TCP	66	443 → 64625 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
3	0.000441	192.168.190.1	192.168.190.130	TCP	54	64625 → 443 [ACK] Seq=1 Ack=1 Win=1051136 Len=0
4	0.029237	192.168.190.1	192.168.190.130	TLSv1.2	190	Client Hello
5	0.029460	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [ACK] Seq=1 Ack=137 Win=30336 Len=0
6	0.029771	192.168.190.130	192.168.190.1	TLSv1.2	1422	Server Hello, Certificate, Server Hello Done
7	0.031355	192.168.190.1	192.168.190.130	TLSv1.2	656	Client Key Exchange, Change Cipher Spec, Finished
8	0.037622	192.168.190.130	192.168.190.1	TLSv1.2	133	Change Cipher Spec, Finished
9	0.038278	192.168.190.1	192.168.190.130	HTTP	175	GET / HTTP/1.1
10	0.038675	192.168.190.130	192.168.190.1	HTTP	655	HTTP/1.1 200 OK (text/html)Continuation
11	0.038859	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [FIN, ACK] Seq=2049 Ack=860 Win=31488 Len=0
12	0.041147	192.168.190.1	192.168.190.130	TCP	54	64625 → 443 [ACK] Seq=860 Ack=2050 Win=1051136 Len=0
13	0.047766	192.168.190.1	192.168.190.130	TLSv1.2	111	Alert (Level: Warning, Description: Close Notify)
14	0.047915	192.168.190.1	192.168.190.130	TCP	54	64625 → 443 [FIN, ACK] Seq=917 Ack=2050 Win=1051136 Len=0
15	0.047919	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [RST] Seq=2050 Win=0 Len=0
16	0.048003	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [RST] Seq=2050 Win=0 Len=0

No.	Time	Source	Destination	Protocol	Length	Info
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3	0.000441	192.168.190.1	192.168.190.130	TCP	54	64625 → 443 [ACK] Seq=1 Ack=1 Win=1051136 Len=0
4	0.029237	192.168.190.1	192.168.190.130	TLSv1.2	190	Client Hello
5	0.029460	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [ACK] Seq=1 Ack=137 Win=30336 Len=0
6	0.029771	192.168.190.130	192.168.190.1	TLSv1.2	1422	Server Hello, Certificate, Server Hello Done
7	0.031385	192.168.190.1	192.168.190.130	TLSv1.2	656	Client Key Exchange, Change Cipher Spec, Finished
8	0.037622	192.168.190.130	192.168.190.1	TLSv1.2	133	Change Cipher Spec, Finished
9	0.038675	192.168.190.1	192.168.190.130	HTTP	175	GET / HTTP/1.1
10	0.038859	192.168.190.130	192.168.190.1	HTTP	655	HTTP/1.1 200 OK (text/html)Continuation
11	0.038859	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [FIN, ACK] Seq=2049 Ack=860 Win=31488 Len=0
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13	0.047766	192.168.190.1	192.168.190.130	TLSv1.2	111	Alert (Level: Warning, Description: Close Notify)
14	0.047915	192.168.190.1	192.168.190.130	TCP	54	64625 → 443 [FIN, ACK] Seq=917 Ack=2050 Win=1051136 Len=0
15	0.047919	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [RST] Seq=2050 Win=0 Len=0
16	0.048003	192.168.190.130	192.168.190.1	TCP	60	443 → 64625 [RST] Seq=2050 Win=0 Len=0

Etag: "59652-cfd-edc33a50bfec6"\r\n	0100	21 44 4f 43 54 59 50 45	20 48 54 4d 4c 20 50 55	IDOCTYPE HTML PU
Accept-Ranges: bytes\r\n	0110	42 4c 49 43 20 22 2d 2f	2f 57 33 43 2f 2f 44 54	BLIC "-/W3C//DT
Content-Length: 285\r\n	0120	44 20 48 54 4d 4c 20 34	2e 30 31 20 54 72 61 6e	D HTML 4.01 Tran
Connection: close\r\n	0130	73 69 74 69 6f 6e 61 6c	2f 2f 45 4e 22 3e 0d 0a	sitional //EN">..
Content-Type: text/html; charset=UTF-8\r\n	0140	3c 6c 69 6e 6b 20 72 65	6c 3d 22 69 63 6f 6e 22	<link rel="icon"
\r\n	0150	20 74 79 70 65 3d 22 69	6d 61 67 65 2f 70 6e 67	types="image/png
[Request in frame: 0]	0160	22 20 68 72 65 66 3d 22	66 61 76 69 63 6f 6e 2e	" href=" favicon.
[Time since request: 397.000 microseconds]	0170	70 6e 67 22 2f 3e 0d 0a	3c 68 74 6d 6c 3e 0d 0a	png"/>.. <html>..
[Request URI: /]	0180	20 3c 68 65 61 64 3e 0d	0a 20 20 20 20 3c 74 69	<head>.. <ti
[Full request URI: https://192.168.190.130/]	0190	74 6c 65 3e 48 6f 6d 65	3c 2f 74 69 74 6c 65 3e	tle>Home </title>
File Data: 285 bytes	01a0	0d 0a 20 20 20 3c 6d 65	74 61 20 68 74 74 70 2d	<meta http-
Line-based text data: text/html (9 lines)	01b0	65 71 75 69 76 3d 2d 43	6f 6e 74 65 6e 74 2d 54	equiv="Content-T
Hypertext Transfer Protocol	01c0	79 70 65 22 20 63 6f 6e	74 65 6e 74 3d 22 74 65	ype" content="te
Excess data after a body (not a new request/response), previous Content-Length bogus?	01d0	78 74 2f 68 74 6d 6c 3b	20 63 68 61 72 73 65 74	xt/html; charset
[Expert Info (Warning/Protocol): Excess data after a body (not a new request/response), previous	01e0	3d 69 73 6f 2d 38 38 35	39 2d 31 22 3e 0d 0a 20	=iso-8859-1">..
File Data: 4 bytes	01f0	20 3c 2f 68 65 61 64 3e	0d 0a 20 3c 62 6f 64 79	</head>.. <body
Data (4 bytes)	0200	3e 57 65 6c 63 6f 6d 65	20 68 6f 6d 65 21 3c 2f	>Welcome home!</
	0210	62 6f 64 79 3e 0d 0a	3c 2f 68 74 6d 6c 3e 0d 0a	body">.. </html>..

Wireshark · Follow HTTP Stream (tcp.stream eq 0) · tls_decrypt.pcapng

GET / HTTP/1.1
Host: 192.168.190.130
User-Agent: curl/7.73.0
Accept: */*

HTTP/1.1 200 OK
Date: Sat, 14 Nov 2020 22:55:00 +0000
Server: Apache
Last-Modified: Wed, 06 Jul 2016 17:51:03 GMT
Etag: "59652-cfd-edc33a50bfec6"
Accept-Ranges: bytes
Content-Length: 285
Connection: close
Content-Type: text/html; charset=UTF-8

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<link rel="icon" type="image/png" href="favicon.png"/>
<html>
<head>
<title>Home</title>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">
</head>
<body>Welcome home!</body>
</html>

Client pkt, 1 server pkt, 1 turn.

Entire conversation (623 bytes) Show as ASCII No delta times Stream 0

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Evidence04.pcapng

During routine monitoring of internal network traffic, analysts discovered suspicious activity between two systems. One host appeared to establish remote access to another workstation and began executing several commands. Your task is to analyze the packet capture to determine what actions were performed and identify details about the systems involved.

1. How many ping requests were sent in the pcap capture?

Echo request and reply in total are 12.

_ws.col.protocol == "ICMP"						
No.	Time	Source	Destination	Protocol	Length	Info
4	12.196170025	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=1/256, ttl=64 (reply in 7)
7	12.196737607	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=1/256, ttl=128 (request in 4)
8	13.209235817	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=2/512, ttl=64 (reply in 9)
9	13.209557789	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=2/512, ttl=128 (request in 8)
10	14.233249758	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=3/768, ttl=64 (reply in 11)
11	14.233768129	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=3/768, ttl=128 (request in 10)
12	15.248605631	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=4/1024, ttl=64 (reply in 13)
13	15.248936891	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=4/1024, ttl=128 (request in 12)
14	16.273077068	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=5/1280, ttl=64 (reply in 15)
15	16.273407477	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=5/1280, ttl=128 (request in 14)
17	17.296626187	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=6/1536, ttl=64 (reply in 19)
19	17.296904299	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=6/1536, ttl=128 (request in 17)

Frame 19: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_4b:e3:60 (08:00:27:4b:e3:60), Dst: PCSSystemtec_7c:8e:8e (08:00:27:7c:8e:8e)
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.22
Internet Control Message Protocol

ICMP traffic includes both requests (type 8) and replies (type 0). By filtering specifically for type 8, you isolate only the "Echo Requests" sent by the source.

icmp.type==8						
No.	Time	Source	Destination	Protocol	Length	Info
4	12.196170025	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=1/256, ttl=64 (reply in 7)
8	13.209235817	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=2/512, ttl=64 (reply in 9)
10	14.233249758	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=3/768, ttl=64 (reply in 11)
12	15.248605631	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=4/1024, ttl=64 (reply in 13)
14	16.273077068	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=5/1280, ttl=64 (reply in 15)
17	17.296626187	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=6/1536, ttl=64 (reply in 19)

2. What is the IP address of the device associated with 08:00:27:4b:e3:60? Ans: 10.0.2.15

eth.addr == 08:00:27:4b:e3:60 , Searching for the MAC address reveals packets where this device is either the source or destination. Examining the IP layer of these packets shows the associated IP.

eth.addr == 08:00:27:4b:e3:60					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000000	10.0.2.15	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
2	3.030682012	10.0.2.15	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
3	6.04689652	10.0.2.15	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
4	12.196170025	10.0.2.22	10.0.2.15	ICMP	98 Echo (ping) request id=0x0841, seq=1/256, ttl=64 (reply in 7)
5	12.196557963	PCSSystemtec_4b:e3:60	Broadcast	ARP	60 Who has 10.0.2.22? Tell 10.0.2.15
6	12.196566149	PCSSystemtec_7c:8e:8e	PCSSystemtec_4b:e3:60	ARP	42 10.0.2.22 is at 08:00:27:7c:8e:8e
7	12.196737607	10.0.2.15	10.0.2.22	ICMP	98 Echo (ping) reply id=0x0841, seq=1/256, ttl=128 (request in 4)
8	13.209235817	10.0.2.22	10.0.2.15	ICMP	98 Echo (ping) request id=0x0841, seq=2/512, ttl=64 (reply in 9)
9	13.209557789	10.0.2.15	10.0.2.22	ICMP	98 Echo (ping) reply id=0x0841, seq=2/512, ttl=128 (request in 8)
10	14.233249758	10.0.2.22	10.0.2.15	ICMP	98 Echo (ping) request id=0x0841, seq=3/768, ttl=64 (reply in 11)
11	14.233768129	10.0.2.15	10.0.2.22	ICMP	98 Echo (ping) reply id=0x0841, seq=3/768, ttl=128 (request in 10)
12	15.248605631	10.0.2.22	10.0.2.15	ICMP	98 Echo (ping) request id=0x0841, seq=4/1024, ttl=64 (reply in 13)
13	15.248936891	10.0.2.15	10.0.2.22	ICMP	98 Echo (ping) reply id=0x0841, seq=4/1024, ttl=128 (request in 12)
14	16.273077068	10.0.2.22	10.0.2.15	ICMP	98 Echo (ping) request id=0x0841, seq=5/1280, ttl=64 (reply in 15)
15	16.273163433	10.0.2.15	10.0.2.22	ICMP	98 Echo (ping) reply id=0x0841, seq=5/1280, ttl=128 (request in 14)

Frame 9: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0 Ethernet II, Src: PCSSystemtec_7c:8e:8e (08:00:27:4b:e3:60), Dst: PCSSystemtec_7c:8e:8e (08:00:27:7c:8e:8e) - Destination: PCSSystemtec_7c:8e:8e (08:00:27:7c:8e:8e) 0. = LG bit: Globally unique address (factory default) 0. = IG bit: Individual address (unicast) - Source: PCSSystemtec_4b:e3:60 (08:00:27:4b:e3:60) 0. = LG bit: Globally unique address (factory default) 0. = IG bit: Individual address (unicast) Type: IPv4 (0x0800) [Stream index: 1] Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.22 0100 = Version: 4 ...0101 = Header Length: 20 bytes (5) Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 84 Identification: 0x00a5 (165) 0000 = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 128 Protocol: ICMP (1) Header Checksum: 0x21e0 [validation disabled] [Header checksum status: Unverified] Source Address: 10.0.2.15 Destination Address: 10.0.2.22 [Stream index: 1]	<pre> 0000 08 00 27 7c 8e 8e 08 00 27 4b e3 60 08 00 45 00 ...K...E... 0010 00 54 00 a5 00 00 00 01 21 e0 0a 00 02 0f 0a 00 ...T...I... 0020 02 16 00 00 9f 05 08 41 00 02 76 86 eb 5e 00 00 ...A...v... 0030 00 00 35 ff 02 00 00 00 00 00 10 11 12 13 14 15 ...S... 0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 ...I*%\$ 0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()*+,-./012345 0060 36 37 67 </pre>
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Ethernet II, Src: PCSSystemtec_4b:e3:60 (08:00:27:4b:e3:60), Dst: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:fa)
  Destination: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:fa)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....1. .... = IG bit: Group address (multicast/broadcast)
  Source: PCSSystemtec_4b:e3:60 (08:00:27:4b:e3:60)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 239.255.255.250

```

3. What version of Internet Group Management Protocol is in use? Ans: Version 3
 Look at the "Type" or "Version" field in the IGMP header of any captured IGMP packet.

igmp					
No.	Time	Source	Destination	Protocol	Length Info
1108	226.987679494	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Leave group 224.0.0.252
1110	226.995032982	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.252 for any sources
1112	226.995788335	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Leave group 224.0.0.252
1114	226.996147126	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.252 for any sources
1133	227.463356104	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.252 for any sources
1503	527.064601259	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Leave group 224.0.0.252
1506	527.071504663	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.252 for any sources
1508	527.073705566	10.0.2.15	224.0.0.22	IGMPv3	60 Membership Report / Leave group 224.0.0.252

4. What is the name of the host located at 10.0.2.22? Ans:

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
439	90.751235246	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
440	90.752764111	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1438	368.017532721	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1439	368.020489865	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1540	636.656977939	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1541	636.659339563	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1583	893.665113499	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1584	893.667053707	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1649	4463.9477553...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x22f0a047
1650	4463.9498758...	0.0.0.2.3	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x22f0a047
1651	4463.9499432...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x22f0a047
1652	4463.9529676...	10.0.2.3	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x22f0a047
1755	4705.8605667...	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0x22f0a047
1756	4705.8649441...	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0x22f0a047

Boot file name not given
Magic cookie: DHCP

- Option: (53) DHCP Message Type (Request)
 - Length: 1
 - DHCP: Request (3)
- Option: (12) Host Name
 - Length: 4
 - Host Name: kali
- Option: (55) Parameter Request List

ip.src=10.0.2.22 and dhcp

No.	Time	Source	Destination	Protocol	Length	Info
439	90.751235246	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
440	90.752764111	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1438	368.017532721	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1439	368.020489865	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1540	636.656977939	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1541	636.659339563	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1583	893.665113499	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1584	893.667053707	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1649	4463.9477553...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x22f0a047
1650	4463.9498758...	0.0.0.2.3	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x22f0a047
1651	4463.9499432...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x22f0a047
1652	4463.9529676...	10.0.2.3	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x22f0a047
1755	4705.8605667...	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0x22f0a047
1756	4705.8649441...	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0x22f0a047

> Frame 439: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface eth0, id 0

> Ethernet II, Src: PCSysntec_7c:8e:8e (08:00:27:7c:8e:8e), Dst: PCSysntec_c1:84:ca (08:00:27:c1:84:ca)

> Internet Protocol Version 4, Src: 10.0.2.22, Dst: 10.0.2.3

> User Datagram Protocol, Src Port: 68, Dst Port: 67

> Dynamic Host Configuration Protocol (Request)

Message type: Boot Request (1)

Hardware type: Ethernet (0x01)

Hardware address length: 6

Hops: 0

Transaction ID: 0xa823310c

Seconds elapsed: 0

Bootp flags: 0x0000 (Unicast)

Client IP address: 10.0.2.22

Your (client) IP address: 0.0.0.0

Next server IP address: 0.0.0.0

Relay agent IP address: 0.0.0.0

Client MAC address: PCSysntec_7c:8e:8e (08:00:27:7c:8e:8e)

Client hardware address padding: 00000000000000000000

Server host name not given

Boot file name not given

Magic cookie: DHCP

- Option: (53) DHCP Message Type (Request)
- Option: (12) Host Name
 - Length: 4
 - Host Name: kali

5. What is the IP address of the DHCP server? Ans: 10.0.2.3

Locate the DHCP ACK (Acknowledgment) packet. This packet is sent by the server to the client to confirm the IP assignment. The "Server Identifier" field within the DHCP options explicitly lists this IP as the authoritative server.

dhcp						
No.	Time	Source	Destination	Protocol	Length	Info
439	90.751235246	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
440	90.752764111	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1438	368.017532721	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1439	368.020489865	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1540	636.656977939	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1541	636.659339563	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1583	893.665113499	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0xa823310c
1584	893.667053707	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0xa823310c
1640	4463.9477553	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x22f0a047
1650	4463.9498758	10.0.2.3	255.255.255.255	DHCP	590	DHCP Offer - Transaction ID 0x22f0a047
1651	4463.9499432	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x22f0a047
1652	4463.9529676	10.0.2.3	255.255.255.255	DHCP	590	DHCP ACK - Transaction ID 0x22f0a047
1755	4705.8605667	10.0.2.22	10.0.2.3	DHCP	342	DHCP Request - Transaction ID 0x22f0a047
1756	4705.8649441	10.0.2.3	10.0.2.22	DHCP	590	DHCP ACK - Transaction ID 0x22f0a047
<pre> Internet Protocol Version 4, Src: 10.0.2.3, Dst: 10.0.2.22 User Datagram Protocol, Src Port: 67, Dst Port: 68 Dynamic Host Configuration Protocol (ACK) Message type: Boot Reply (2) Hardware type: Ethernet (0x01) Hardware address length: 6 Hops: 0 Transaction ID: 0xa823310c Seconds elapsed: 0 Bootp flags: 0x0000 (Unicast) Client IP address: 10.0.2.22 Your (client) IP address: 10.0.2.22 Next server IP address: 0.0.0.0 Relay agent IP address: 0.0.0.0 Client MAC address: PCSSystemtec 7c:8e:8e (08:00:27:7c:8e:8e) Client hardware address padding: 00000000000000000000 Server host name not given Boot file name not given Magic cookie: DHCP Option: (54) DHCP Server Identifier (10.0.2.3) Length: 4 DHCP Server Identifier: 10.0.2.3 </pre>						

6. What is the name of the host located at 10.0.2.15? Ans: Workgroup is a group of hosts, it can't be the host name.

Ip addr and smb and nbns check

smb						
No.	Time	Source	Destination	Protocol	Length	Info
1478	385.290370566	10.0.2.15	10.0.2.255	BROWSER	243	Host Announcement MSEDGWIN10, Workstation, Server, NT Workstation
1623	4448.2081508	10.0.2.15	10.0.2.255	BROWSER	243	Host Announcement MSEDGWIN10, Workstation, Server, NT Workstation
2681	5170.7114308	10.0.2.15	10.0.2.255	BROWSER	243	Host Announcement MSEDGWIN10, Workstation, Server, NT Workstation
2914	5674.8862001	10.0.2.15	10.0.2.255	BROWSER	216	Get Backup List Request
2918	5678.1744447	10.0.2.15	10.0.2.255	BROWSER	216	Get Backup List Request
2926	5681.4705337	10.0.2.15	10.0.2.255	BROWSER	216	Get Backup List Request
3308	5892.6651734	10.0.2.15	10.0.2.255	BROWSER	243	Host Announcement MSEDGWIN10, Workstation, Server, NT Workstation
6799	6430.8796610	10.0.2.15	10.0.2.255	BROWSER	243	Host Announcement MSEDGWIN10, Workstation, NT Workstation
6891	6485.0571109	10.0.2.15	10.0.2.255	BROWSER	243	Host Announcement MSEDGWIN10, Workstation, Server, NT Workstation
<pre> Frame 1478: Packet, 243 bytes on wire (1944 bits), 243 bytes captured (1944 bits) on interface eth0, id 0 Ethernet II, Src: PCSSystemtec 4b:e3:60 (08:00:27:4b:e3:60), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.255 User Datagram Protocol, Src Port: 138, Dst Port: 138 NetBIOS Datagram Service SMB (Server Message Block Protocol), Trans Request (0x25) SMB Mailslot Protocol Microsoft Windows Browser Protocol Command: Host Announcement (0x01) Update Count: 0 Update Periodicity: 12 minutes Host Name: MSEDGWIN10 Windows version: OS Major Version: 10 OS Minor Version: 0 Server Type: 0x00001003, Workstation, Server, NT Workstation </pre>						

7. What is the IP address of the attacker? Ans: 10.0.2.15

Analyze protocols like RDP, TCP, and ICMP. Use Statistics > Conversations > IPv4 to see which IP is initiating the most connections. By following TCP Conversations, we observed multiple connection attempts and command executions originating from a single host. Attackers often use one compromised or external host to scan or "pivot" to others; here, the volume and type of traffic (like RDP and ICMP) from .22 indicated malicious intent.

Wireshark - Conversations - evidence04.pcapng

Conversation Settings

☐ Name resolution

☐ Absolute start time

☐ Display raw data

☐ Limit to display filter

Copy

Ethernet - 20

IPv4 - 10

IPv6 - 9

TCP - 22

UDP - 56

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
0.0.0.0	255.255.255.255	4	1 kB	8	4	1 kB	0	0 bytes	4463.947755379	2014.405128	5 bits/s	0 bits/s
10.0.2.3	255.255.255.255	4	2 kB	9	4	2 kB	0	0 bytes	4463.949875891	2014.405991	9 bits/s	0 bits/s
10.0.2.15	10.0.2.255	65	9 kB	7	65	9 kB	0	0 bytes	385.290370566	6769.483882	10 bits/s	0 bits/s
10.0.2.15	224.0.0.22	76	5 kB	4	76	5 kB	0	0 bytes	226.987679494	6852.209439	5 bits/s	0 bits/s
10.0.2.15	224.0.0.251	65	6 kB	5	65	6 kB	0	0 bytes	226.996790320	6852.081195	6 bits/s	0 bits/s
10.0.2.15	224.0.0.252	23	2 kB	6	23	2 kB	0	0 bytes	226.997409055	6852.080040	1 bits/s	0 bits/s
10.0.2.15	239.255.255.250	93	23 kB	0	93	23 kB	0	0 bytes	0.000000000	7156.103063	25 bits/s	0 bits/s
10.0.2.22	10.0.2.3	28	13 kB	3	14	5 kB	14	8 kB	90.751235246	7065.423553	5 bits/s	9 bits/s
10.0.2.22	10.0.2.15	13,192	6 MB	1	7,673	854 kB	5,519	5 MB	12.196170025	7151.453319	955 bits/s	5546 bits/s
10.0.2.22	103.86.96.100	4	428 bytes	2	2	164 bytes	2	264 bytes	76.120909782	6554.761955	0 bits/s	0 bits/s

In the conversation we found only 2 IPs. Go to TCP and check packets transferred

Wireshark - Conversations - evidence04.pcapng

Conversation Settings

☐ Name resolution

☐ Absolute start time

☐ Display raw data

☐ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

☐ Bluetooth

☐ BPv7

☐ DCCP

☐ DNP 3.0

☒ Ethernet

Ethernet - 20

IPv4 - 10

IPv6 - 9

TCP - 22

UDP - 56

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	Flows
10.0.2.22	38620	10.0.2.15	3389	6,786	4 MB	9	3,959	445 kB	2,827	3 MB	6630.883554835	532.765934	6675 bits/s	51 kbps	1,545
10.0.2.22	38604	10.0.2.15	3389	6,104	2 MB	1	3,552	399 kB	2,552	1 MB	76.153720226	6354.712179	502 bits/s	1859 bits/s	1,349
10.0.2.22	39030	10.0.2.15	8888	215	20 kB	7	114	6 kB	101	14 kB	5151.944408458	878.337539	57 bits/s	124 bits/s	29
10.0.2.22	38602	10.0.2.15	3389	16	3 kB	0	10	911 bytes	6	2 kB	76.07689022	0.077148	94 kbps	168 kbps	7
10.0.2.22	38618	10.0.2.15	3389	16	3 kB	8	10	911 bytes	6	2 kB	6630.801360508	0.082603	88 kbps	157 kbps	7
10.0.2.22	42792	10.0.2.15	9999	11	663 bytes	2	6	357 bytes	5	306 bytes	265.654751295	4218.001778	0 bits/s	0 bits/s	1
10.0.2.15	1029	10.0.2.22	8888	2	120 bytes	4	1	66 bytes	1	54 bytes	4819.215037265	0.000021			0
10.0.2.15	1029	10.0.2.22	8888	2	120 bytes	5	1	66 bytes	1	54 bytes	4819.719897976	0.000019			0
10.0.2.15	1029	10.0.2.22	8888	2	120 bytes	6	1	66 bytes	1	54 bytes	4820.220569226	0.000020			0
10.0.2.15	49761	10.0.2.22	31337	2	120 bytes	10	1	66 bytes	1	54 bytes	6922.715676147	0.000022			0
10.0.2.15	49761	10.0.2.22	31337	2	120 bytes	11	1	66 bytes	1	54 bytes	6923.222823452	0.000020			0
10.0.2.15	49761	10.0.2.22	31337	2	120 bytes	12	1	66 bytes	1	54 bytes	6923.753349332	0.000022			0
10.0.2.15	49764	10.0.2.22	31337	2	120 bytes	13	1	66 bytes	1	54 bytes	6964.897414955	0.000073			0
10.0.2.15	49764	10.0.2.22	31337	2	120 bytes	14	1	66 bytes	1	54 bytes	6965.424687692	0.000023			0
10.0.2.15	49764	10.0.2.22	31337	2	120 bytes	15	1	66 bytes	1	54 bytes	6965.925477451	0.000024			0
10.0.2.15	49765	10.0.2.22	31337	2	120 bytes	16	1	66 bytes	1	54 bytes	6971.299700185	0.000026			0
10.0.2.15	49765	10.0.2.22	31337	2	120 bytes	17	1	66 bytes	1	54 bytes	6971.815102375	0.000019			0
10.0.2.15	49765	10.0.2.22	31337	2	120 bytes	18	1	66 bytes	1	54 bytes	6972.327181110	0.000019			0
10.0.2.15	49768	10.0.2.22	31337	2	120 bytes	19	1	66 bytes	1	54 bytes	7037.192548482	0.000021			0
10.0.2.15	49768	10.0.2.22	31337	2	120 bytes	20	1	66 bytes	1	54 bytes	7037.698781718	0.000022			0
10.0.2.15	49768	10.0.2.22	31337	2	120 bytes	21	1	66 bytes	1	54 bytes	7038.213239100	0.000020			0
10.0.2.22	39028	10.0.2.15	8888	2	134 bytes	3	1	74 bytes	1	60 bytes	4662.410976231	0.000271			0

Check all the streams for confirmation. By the help of stream ID we got to know that few are encrypted and one is for testing but in stream 7 we got a plain text file to analyse.

Wireshark - Conversations - evidence04.pcapng

Conversation Settings

- ☐ Name resolution
- ☐ Absolute start time
- ☐ Display raw data
- ☐ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

- ☐ Bluetooth
- ☐ BPv7
- ☐ DCCP
- ☐ DNP 3.0
- ☒ Ethernet
- ☐ FC
- ☐ FDDI
- ☐ IEEE 802.11
- ☐ IEEE 802.15.4

Wireshark - Follow TCP Stream (tcp.stream eq 7) - evidence04.pcapng

```

10.0.2.22 → 10.0.2.15: Microsoft Windows [Version 10.0.17763.1282]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\IEUser> whoami
whoami
msedgewin10\ieuser

C:\Users\IEUser> netstat -ano
netstat -ano

Active Connections
Proto Local Address Foreign Address State PID
TCP 0.0.0.0:135 0.0.0.0:0 LISTENING 872
TCP 0.0.0.0:445 0.0.0.0:0 LISTENING 4
TCP 0.0.0.0:3389 0.0.0.0:0 LISTENING 552
TCP 0.0.0.0:5040 0.0.0.0:0 LISTENING 4420
TCP 0.0.0.0:5985 0.0.0.0:0 LISTENING 4
TCP 0.0.0.0:47001 0.0.0.0:0 LISTENING 4
TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING 480

```

icmp

No.	icmp	Source	Destination	Protocol	Length	Info
7	icmpv6	12.196737607	10.0.2.22	ICMP	98	Echo (ping) request id=0x0841, seq=1/256, ttl=64 (reply in 7)
8	icmpv6	13.209235817	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=1/256, ttl=128 (request in 4)
9	icmpv6	13.209557789	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=2/512, ttl=64 (reply in 9)
10	icmpv6	14.233249758	10.0.2.15	ICMP	98	Echo (ping) reply id=0x0841, seq=2/512, ttl=128 (request in 8)
	icmpv6			ICMP	98	Echo (ping) request id=0x0841, seq=3/768, ttl=64 (reply in 11)

No.	Time	Source	Destination	Protocol	Length	Info
29	76.077285472	10.0.2.22	10.0.2.15	RDP	98	Cookie: msthash=IEUser, Negotiate Request
30	76.111127623	10.0.2.15	10.0.2.22	RDP	73	Negotiate Response
48	76.153963697	10.0.2.22	10.0.2.15	RDP	98	Cookie: msthash=IEUser, Negotiate Request
50	76.159133419	10.0.2.15	10.0.2.22	RDP	73	Negotiate Response
6999	6630.8034697...	10.0.2.22	10.0.2.15	RDP	98	Cookie: msthash=IEUser, Negotiate Request
7000	6630.8457095...	10.0.2.15	10.0.2.22	RDP	73	Negotiate Response
7018	6630.8838801...	10.0.2.22	10.0.2.15	RDP	98	Cookie: msthash=IEUser, Negotiate Request
7020	6630.8893907...	10.0.2.15	10.0.2.22	RDP	73	Negotiate Response

No.	Time	Source	Destination	Protocol	Length	Info
26	76.076899022	10.0.2.22	10.0.2.15	TCP	74	38602 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2365737254 TSecr=0 WS=128
27	76.077200871	10.0.2.15	10.0.2.22	TCP	66	3389 → 38602 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
28	76.077217570	10.0.2.22	10.0.2.15	TCP	54	38602 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0
29	76.077285472	10.0.2.22	10.0.2.15	RDP	98	Cookie: msthash=IEUser, Negotiate Request
30	76.111127623	10.0.2.15	10.0.2.22	RDP	73	Negotiate Response
31	76.111154001	10.0.2.22	10.0.2.15	TCP	54	38602 → 3389 [ACK] Seq=45 Ack=20 Win=64256 Len=0
32	76.113273387	10.0.2.22	10.0.2.15	TLSv1	158	Client Hello

8. What was the first command run by the attacker? Ans: *whoami*

```
Wireshark · Follow TCP Stream (tcp.stream eq 7) · evidence04.pcapng

Microsoft Windows [Version 10.0.17763.1282]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>
whoami

whoami
msedgewin10\ieuser

C:\Users\IEUser>
netstat -ano

netstat -ano

Active Connections

Proto Local Address Foreign Address State PID
TCP 0.0.0.0:135 0.0.0.0:0 LISTENING 872
```

9. What was the process ID of the RDP session? Ans: 552

```
Wireshark · Follow TCP Stream (tcp.stream eq 7) · evidence04.pcapng

Microsoft Windows [Version 10.0.17763.1282]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>
whoami

whoami
msedgewin10\ieuser

C:\Users\IEUser>
netstat -ano

netstat -ano

Active Connections

Proto Local Address Foreign Address State PID
TCP 0.0.0.0:135 0.0.0.0:0 LISTENING 872
TCP 0.0.0.0:445 0.0.0.0:0 LISTENING 4
TCP 0.0.0.0:3389 0.0.0.0:0 LISTENING 552
TCP 0.0.0.0:5040 0.0.0.0:0 LISTENING 4420
```

10. What was the maximum password age for the account IEUser? Ans: Age: 42

```
C:\Users\IEUser>
net accounts

net accounts
Force user logoff how long after time expires?: Never
Minimum password age (days): 0
Maximum password age (days): 42
Minimum password length: 0
Length of password history maintained: None
Lockout threshold: Never
Lockout duration (minutes): 30
Lockout observation window (minutes): 30
Computer role: WORKSTATION
The command completed successfully.
```

11. What is the file path for the location of working capital analysis.xls? Ans:

C:\Users\IEUser\Documents

```
C:\Users\IEUser\Documents>
dir

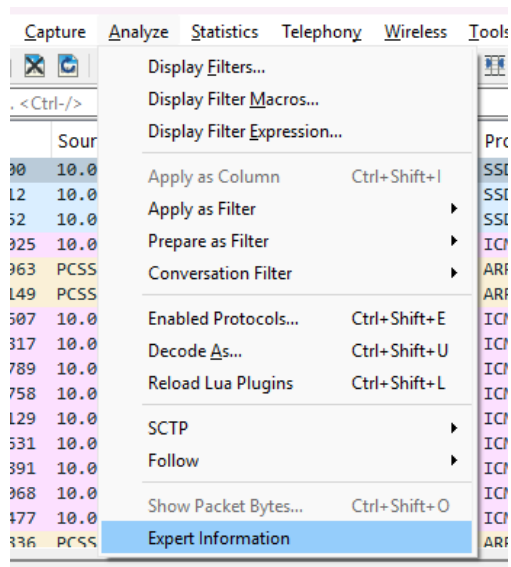
dir
Volume in drive C is Windows 10
Volume Serial Number is B009-E7A9

Directory of C:\Users\IEUser\Documents

06/18/2020  07:50 AM    <DIR>          .
06/18/2020  07:50 AM    <DIR>          ..
06/18/2020  07:38 AM             153,088 balance sheet 2017.xls
06/18/2020  07:43 AM             1,925,632 balance sheet assets.xls
06/18/2020  07:45 AM              203,776 five year summary financial data.xls
06/18/2020  07:39 AM              398,336 working capital analysis.xls
               4 File(s)          2,680,832 bytes
               2 Dir(s)  20,073,222,144 bytes free

C:\Users\IEUser\Documents>
```

12. What is the message hidden in the comments? Ans: Hidden comment is “pcap analysis is fun!”



Wireshark · Expert Information · evidence04.pcapng

Packet	Summary	Group	Protocol	Count
> Warning	DNS query retransmission	Protocol	LLMNR	8
> Warning	DNS query retransmission	Protocol	mDNS	18
> Warning	DNS response missing	Protocol	mDNS	4
> Warning	DNS response missing	Protocol	LLMNR	39
> Warning	Connection reset (RST)	Sequence	TCP	19
> Note	A new tcp session is started with the same ports as an earlier session in th...	Sequence	TCP	10
> Note	This frame undergoes the connection closing	Sequence	TCP	2
> Note	Time To Live	Sequence	IPv4	65
> Note	This packet's length exceeds MSS (common with TSO or incomplete con...	Protocol	TCP	552
> Note	This frame initiates the connection closing	Sequence	TCP	5
> Chat	Connection finish (FIN)	Sequence	TCP	7
> Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP	6
> Chat	Connection establish request (SYN)	Sequence	TCP	22
▼ Comment	Packet comments listed below.	Comment	Packet	1
1	flag: pcap analysis is fun!	Comment	Packet	

Apply a display filter == <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2	3.03062012	10.0.2.15	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
3	6.046899652	10.0.2.15	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
4	12.196170025	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=1/256, ttl=64 (reply in 7)
5	12.196557963	PCSSystemtec_4b:e3:: Broadcast		ARP	60	Who has 10.0.2.22? Tell 10.0.2.15
6	12.196566149	PCSSystemtec_7c:8e:: PCSSystemtec_4b:e3::		ARP	42	10.0.2.22 is at 08:00:27:7c:8e:8e
7	12.196737607	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=1/256, ttl=128 (request in 4)
8	13.209235817	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=2/512, ttl=64 (reply in 9)
9	13.209557789	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=2/512, ttl=128 (request in 8)
10	14.233249758	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=3/768, ttl=64 (reply in 11)
11	14.233768129	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=3/768, ttl=128 (request in 10)
12	15.24805631	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=4/1024, ttl=64 (reply in 13)
13	15.248936891	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=4/1024, ttl=128 (request in 12)
14	16.273077068	10.0.2.22	10.0.2.15	ICMP	98	Echo (ping) request id=0x0841, seq=5/1280, ttl=64 (reply in 15)
15	16.273407477	10.0.2.15	10.0.2.22	ICMP	98	Echo (ping) reply id=0x0841, seq=5/1280, ttl=128 (request in 14)
16	17.296570836	PCSSystemtec_7c:8e:: PCSSystemtec_4b:e3::		ARP	42	Who has 10.0.2.15? Tell 10.0.2.22

▼ Packet comments

flag: pcap analysis is fun!

> Frame 1: Packet: 179 bytes on wire (1432 bits): 179 bytes captured (1432 bits) on interface eth0, id 0

```

0000  01 00 5e 7f ff fa 08 00 27 4b e3 60 08 00 45 00  ...A...K...E
0010  00 a5 3b ac 00 00 04 11 7e 93 0a 00 02 0f ef ff  ...;.....
0020  ff fa d5 50 07 6c 00 91 33 e8 4d 2d 53 45 41 52  ...P...3-M-SEAR

```